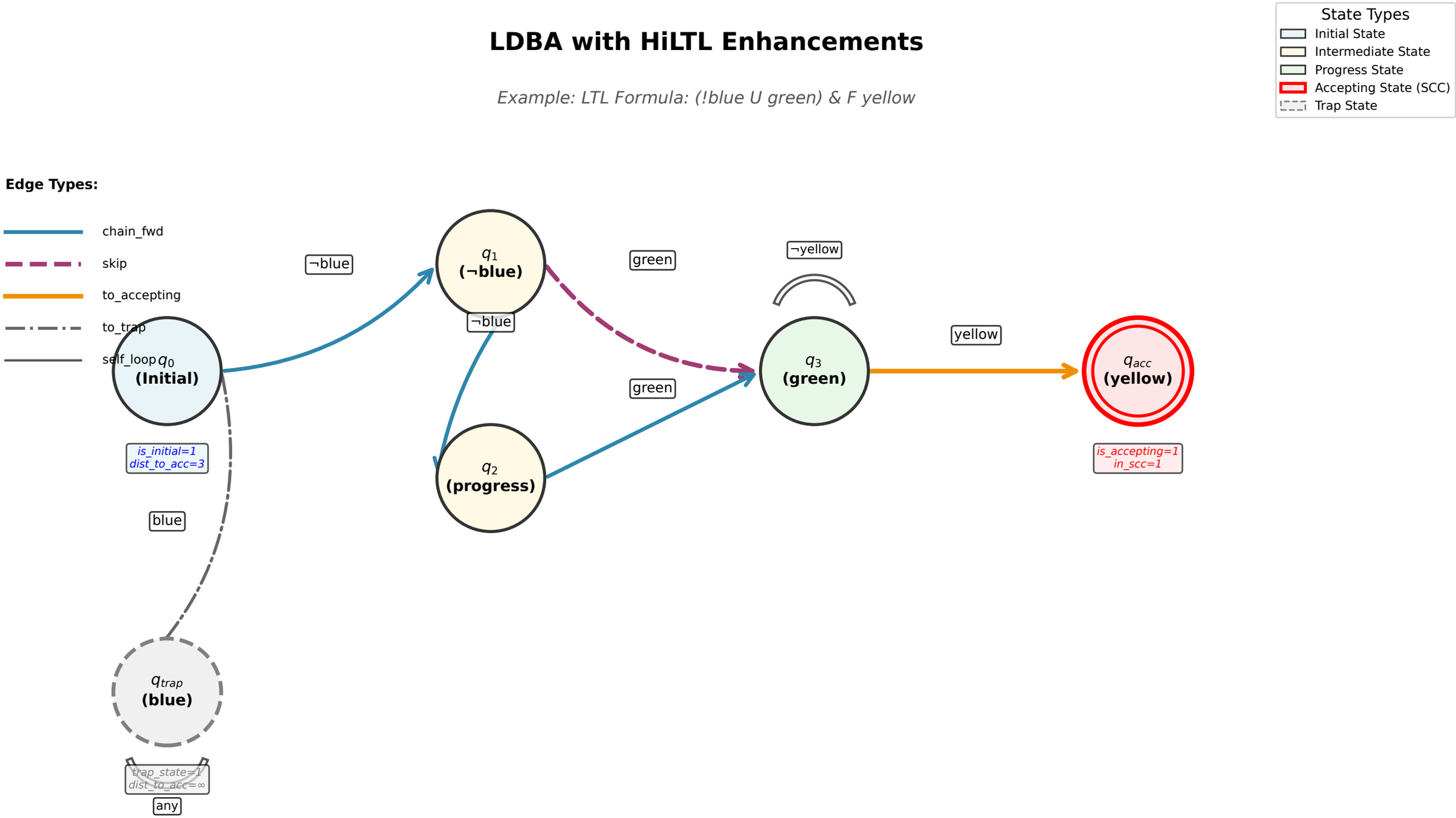
LDBA with HiLTL Enhancements

Example: LTL Formula: (!blue U green) & F yellow



① SE-GNN Enhancement

• Edge features (6D):

type + semantic distance

• Node features (4D):

accepting + SCC + distance

② Multi-Scale GNN

• Layer 0: local (1-hop)

• Layer 1: mid-range (2-hop)

• Layer 2: global

→ Short/Mid/Long embeddings

③ Hierarchical Value

• V_short(s): env state

• V_long(q): automaton state

• α(q): dynamic fusion

V = α·V_long + (1-α)·V_short